



Introduction

Large language models excel at generating fluent text but rely on limited, static knowledge. RAG grounds outputs in external evidence, yet struggles with **retrieval noise** and **spurious correlations** in complex tasks like HotpotQA.

This project:

- Investigates how reasoning-based fine-tuning improves RAG’s evidence usage
- Compares **No RAG** (backbone model only), **Vanilla RAG**, **Self-RAG**, and **InstructRAG**
- Evaluates how different supervision signals affect reasoning quality

Vanilla RAG

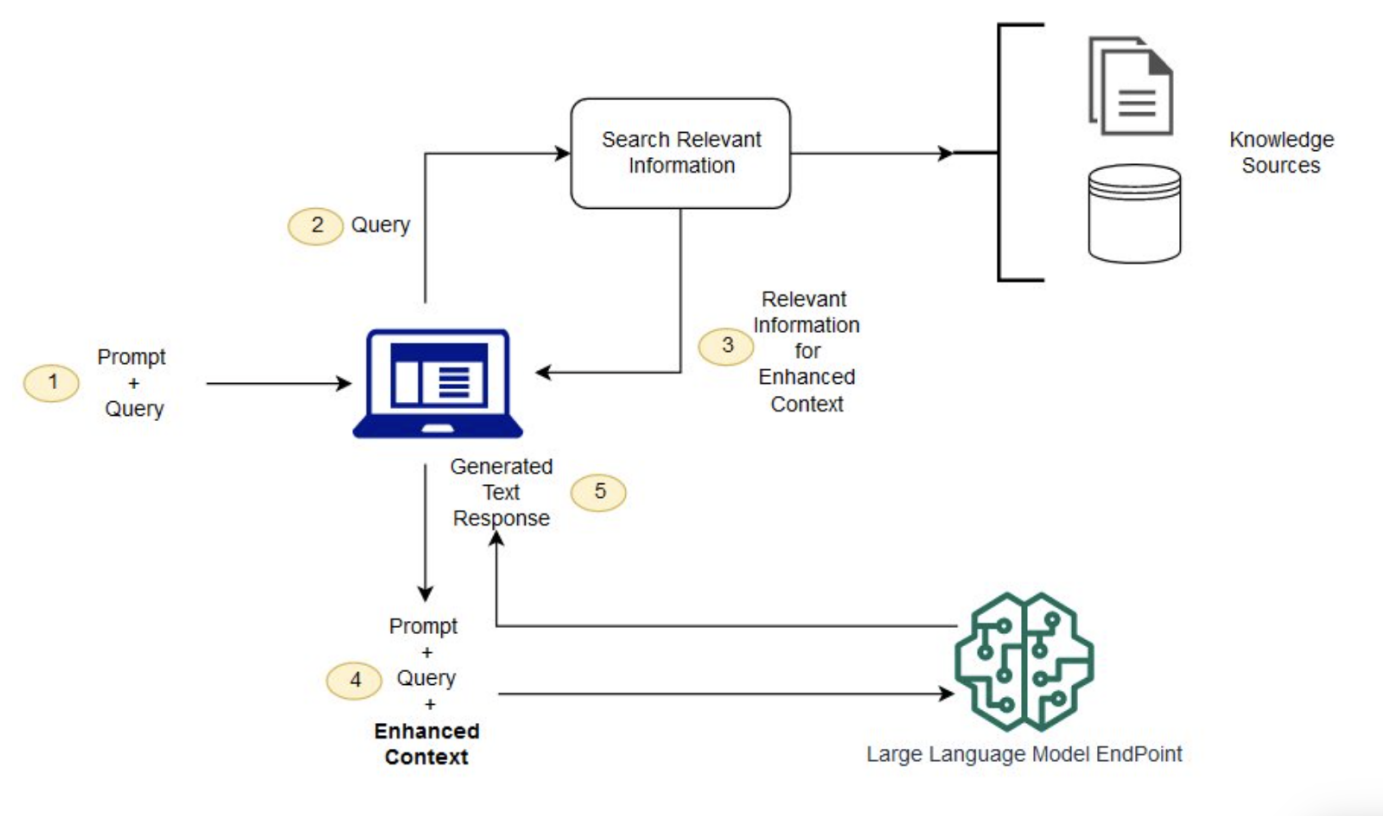


Figure 1. Standard RAG pipeline: retrieve-then-generate architecture. (Source: AWS, 2024)

RAG Variations

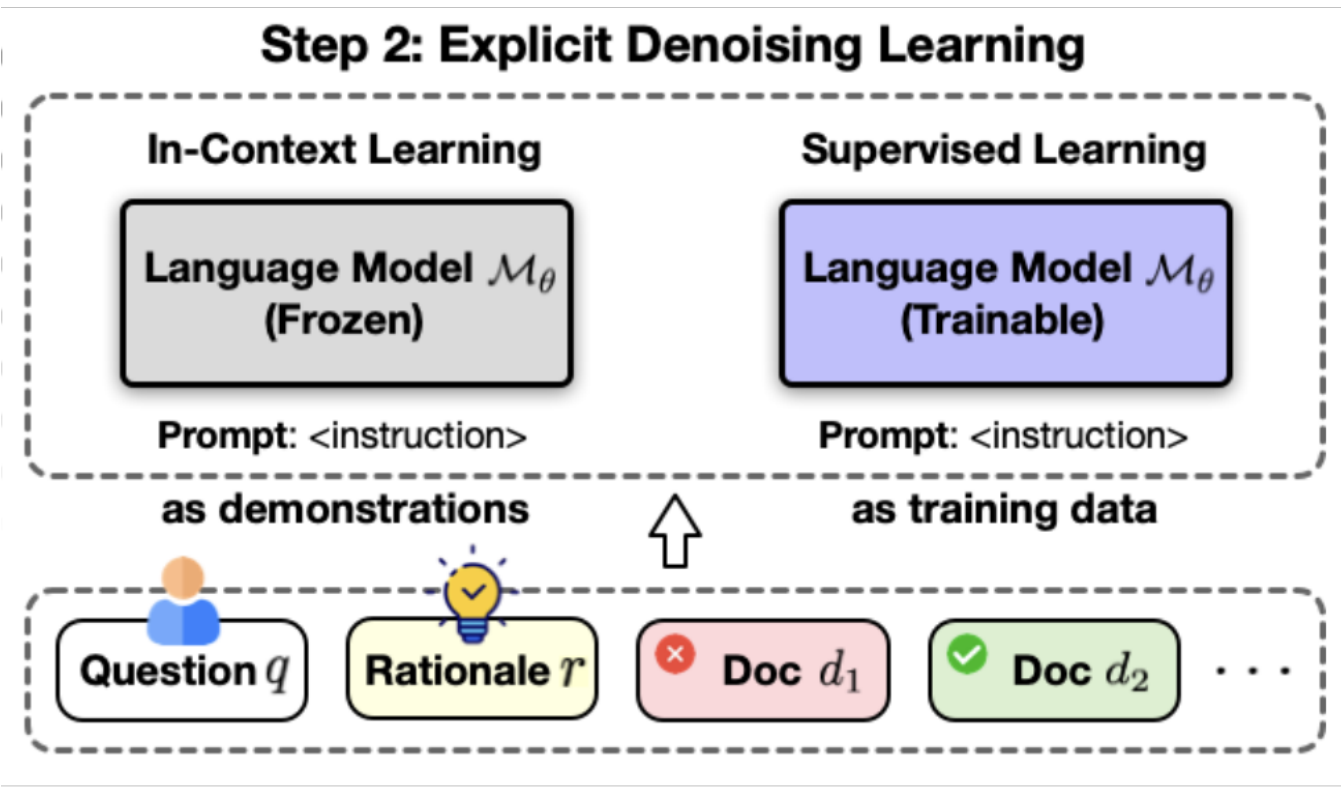
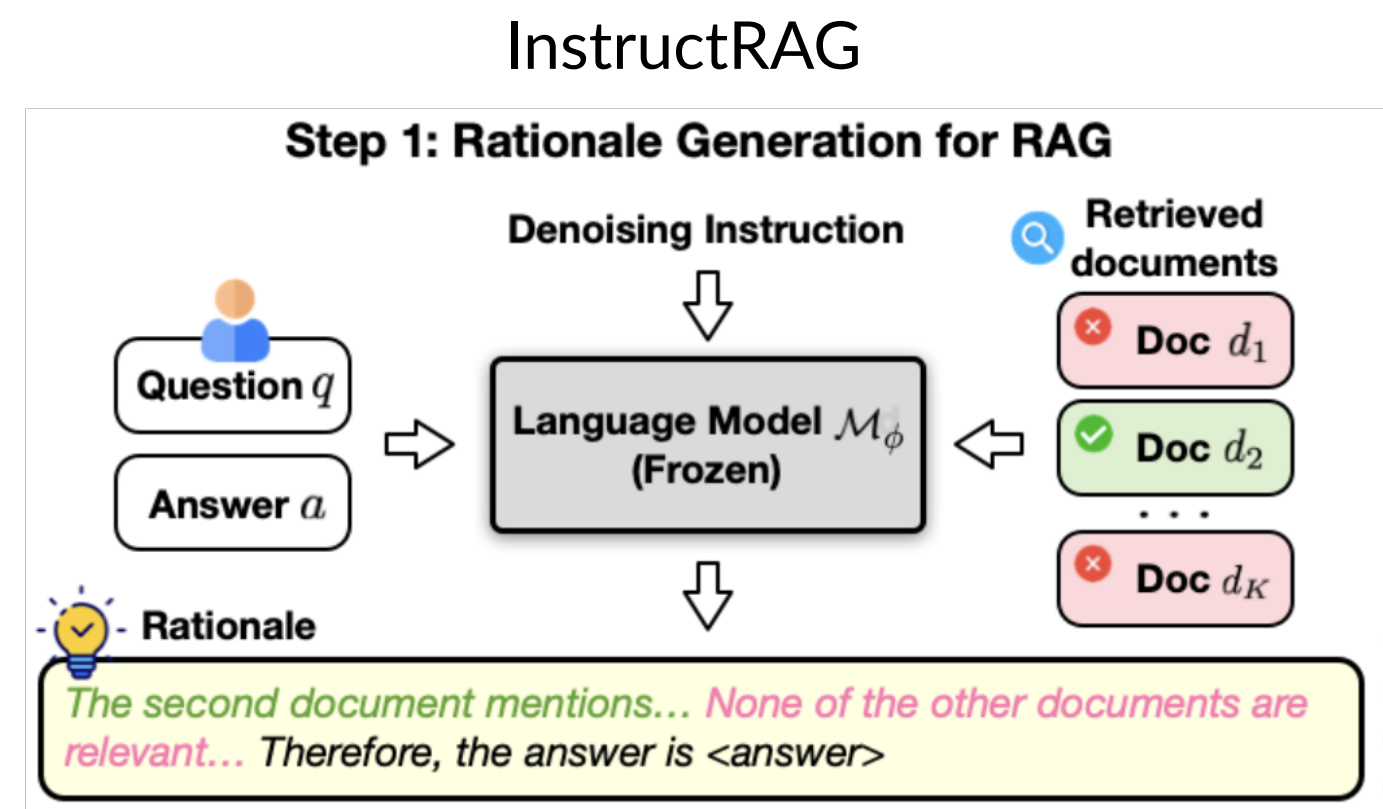


Figure 2. Linking documents via multi-hop reasoning chains (Source: Wei et al, 2024).

Self-RAG

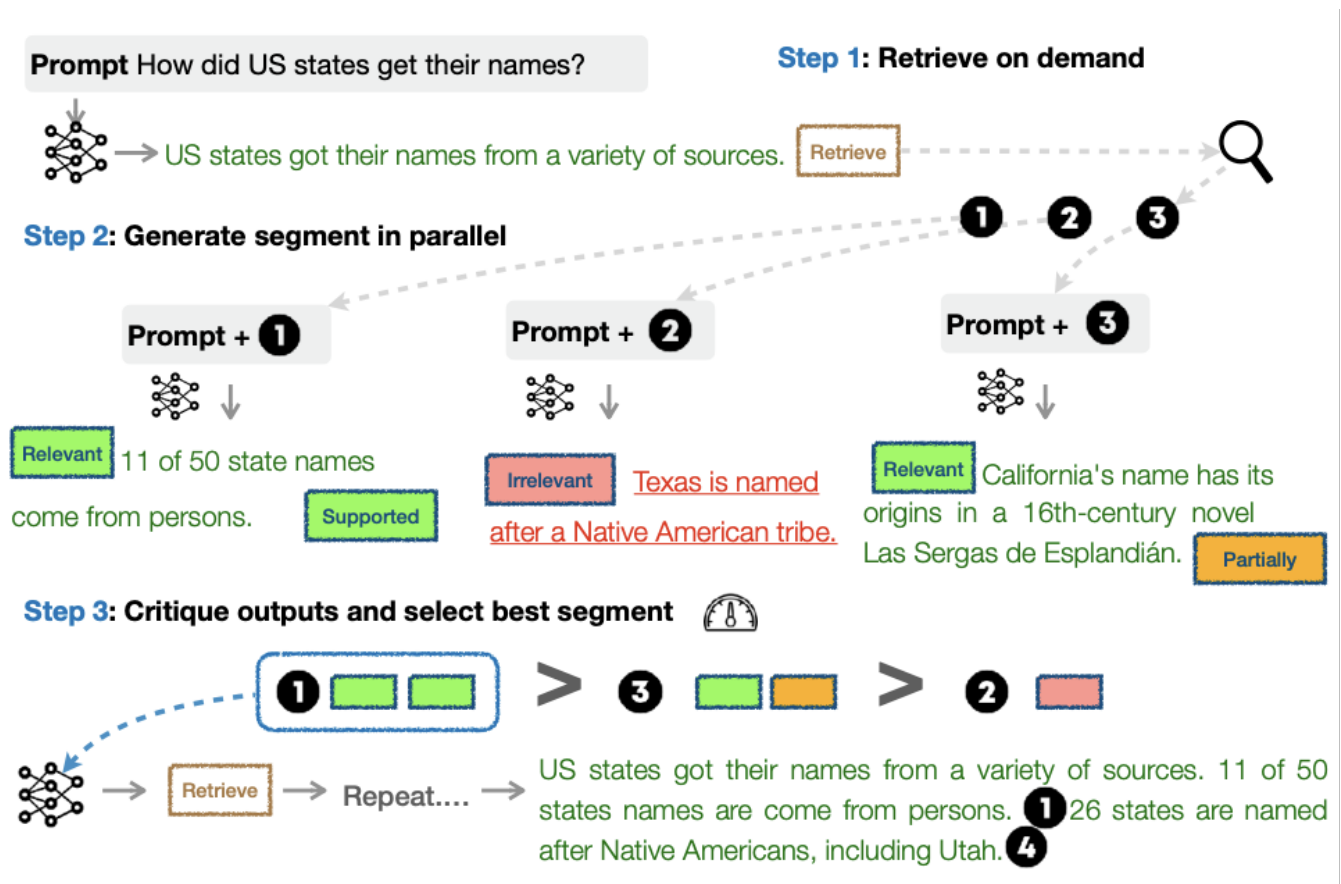


Figure 3. Generates, critiques, and selects the best-supported response segments through reflection (Source: Asai et al, 2023).

Methods

Backbone Model:

- Llama-2-7B** (7B parameters, 2T tokens)
- Base pretrained version without fine-tuning to isolate RAG strategy impact
- Retriever:** Contriever (top-K=5-10 Wikipedia passages)

Evaluation Metrics:

- Exact Match (EM):** Binary correctness after normalization
- F1 Score:** Token-level overlap via precision and recall
- Semantic Similarity:** Cosine similarity using sentence embeddings
- Accuracy:** Whether any gold answer appears in prediction

Dataset:

- HotPotQA** (Yang et al., 2018) development set (7,405 examples)
- Multi-hop reasoning across documents
- Provides diversity with gold annotations while remaining computationally feasible

Results

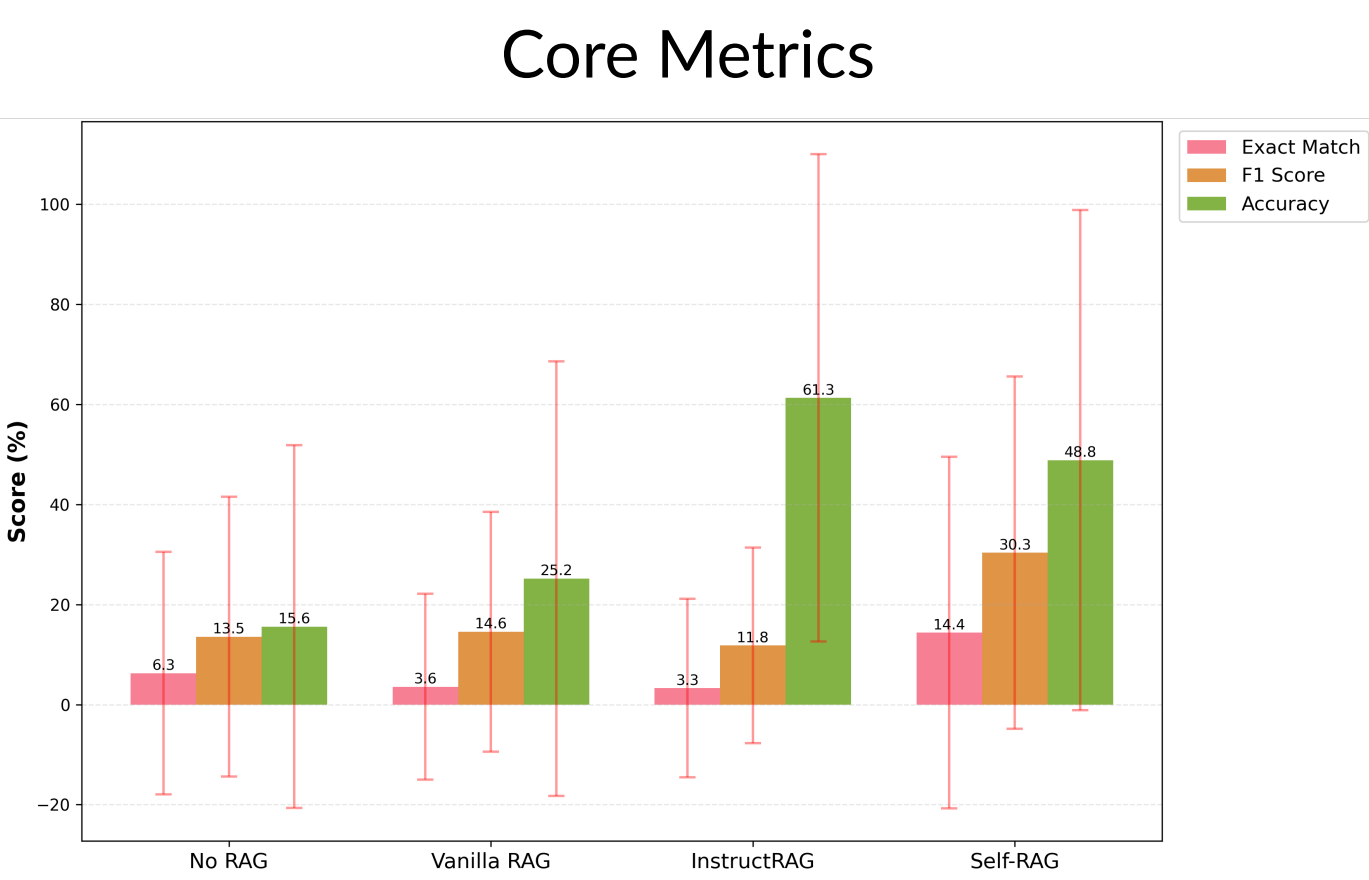


Figure 4. Core evaluation metrics (Accuracy, EM, and F1) of the RAG variations.

Response Length

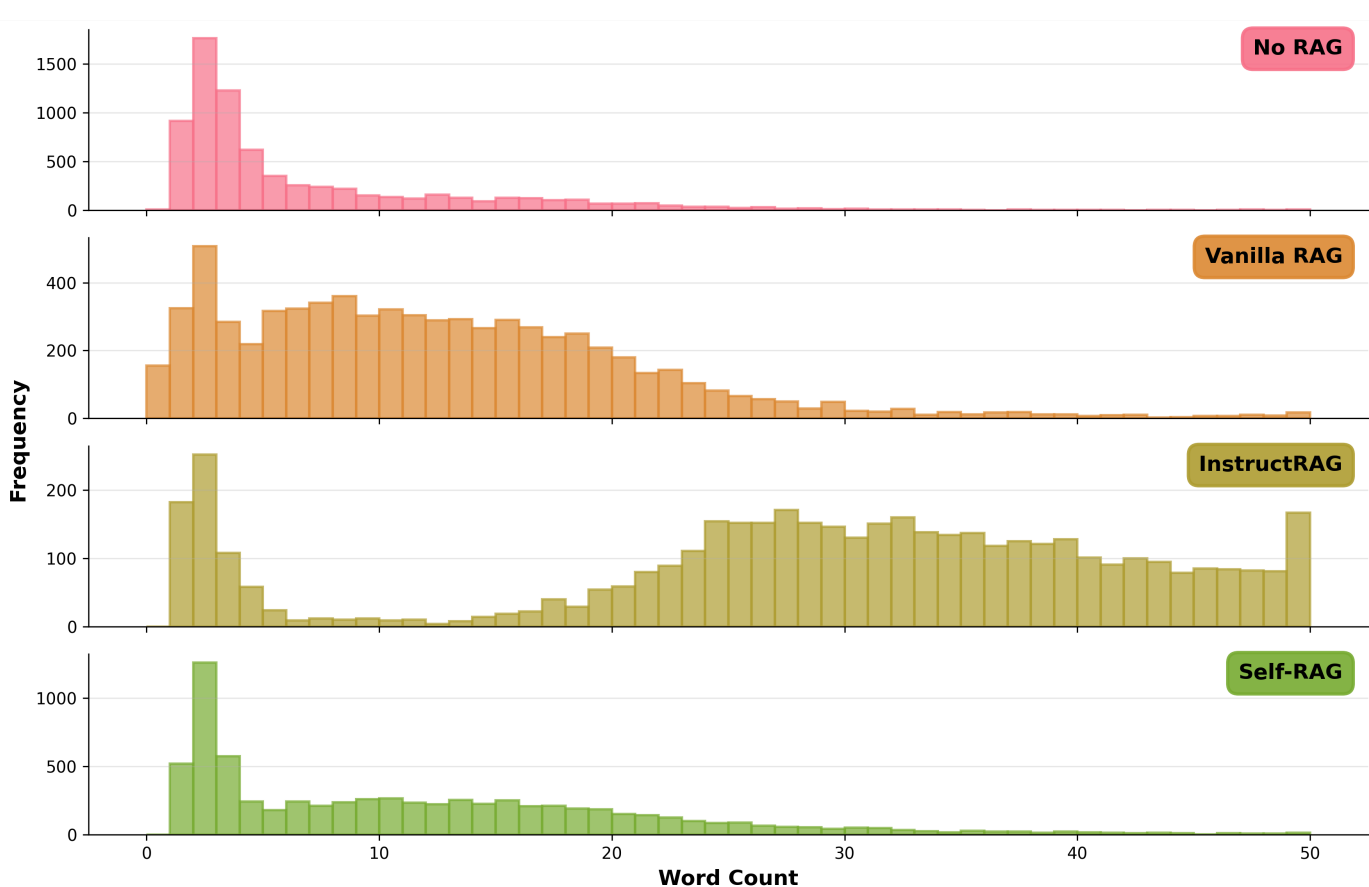


Figure 5. Response length of the different RAG variations.

Semantic Similarity

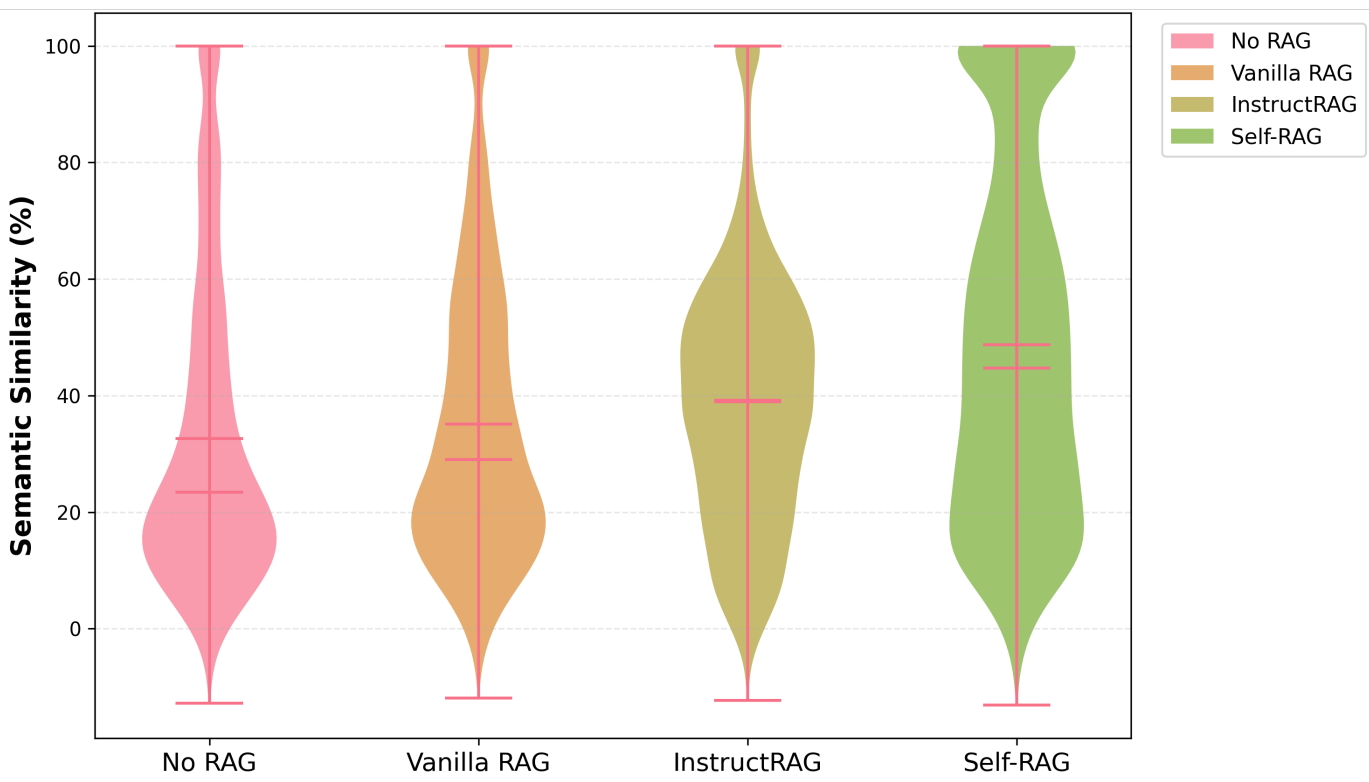


Figure 6. Semantic similarity of each RAG variation's response to the gold answer.

Example Responses

Question: Are Local H and For Against both from the United States?

Gold Answer: [‘yes’]

No RAG Response (Accuracy=0, F1=0.00):
No, they are not.

Hallucinates confidently without factual knowledge, demonstrating parametric memory limitations.

Vanilla RAG Response (Accuracy=1, F1=0.29):
Yes, both are from the United States.

Answers correctly but underutilizes context.

InstructRAG Response (Accuracy=1, F1=0.01):

Yes, based on the information provided in the documents, Local H and For Against are both from the United States.

Document 5 (Title: Local H) states [...]

Similarly, Document 7 (Title: For Against) mentions that [...]

[...] based on the available evidence, it can be concluded that both Local H and For Against are from the United States.

Provides detailed evidence-based reasoning but over-generates (140+ words vs. gold “yes”), severely hurting F1.

Self-RAG Response (Accuracy=1, F1=0.17):

Yes, both Local H and For Against are from the United States.

Best balances correctness and conciseness through calibrated generation.

Conclusion

- Advanced RAG variants (Self-RAG and InstructRAG) improve overall performance in multi-hop QA, as evidenced by **higher semantic similarity** (48.75% and 39.16% vs. 35.11% by Vanilla RAG)
- Response characteristics vary significantly** due to different augmentation mechanisms, resulting in dramatically different response lengths (Self-RAG: 11.8 vs. InstructRAG: 53.9 tokens)
- Simple accuracy metrics are insufficient:** Claims of “state-of-the-art” RAG performance require scrutiny, as evaluation must account for response format, length, and reasoning transparency alongside correctness
- Complexity-accuracy tradeoff** exists: Enhanced performance comes at the cost of increased computational overhead and longer responses

Discussions

Future RAG evaluation frameworks must move beyond single-metric comparisons to holistic assessments that consider reasoning quality, interpretability, and computational efficiency. Additionally, optimal metric choice depends heavily on use case—accuracy for factual lookup, semantic similarity for open-ended generation, and F1 for precise extraction—highlighting the need for context-aware evaluation strategies.